CONDITIONING IN GENERATIVE MODELS

In generative modelling, *conditioning* means that generation is not random, but guided by some external input

- **Unconditional generation**

$$p\,(\text{image})$$

The model learns the overall distribution of natural images, but has no control over specific content

- **Conditional generation**

$$p\,(\text{image} \mid \text{text})$$

The model learns how images vary when the textual description changes

- The text acts as **constraint** that shapes the probability space of possible images
- Different prompts shift the probability mass toward different semantic regions of the image space (objects, scenes, and styles)

TEXT ENCODING

Images are continuous pixel arrays, while text is discrete tokens. Conditioning requires mapping both into compatible numerical representations

Step 1: Text Encoding

A text encoder (usually a Transformer) converts the prompt into embeddings

$$\text{text} = (w_1, w_2, \ldots, w_T) \xrightarrow{\text{Text Encoder}} (\mathrm{e}_1, \mathrm{e}_2, \ldots, \mathrm{e}_T)$$

- Each token becomes an embedding capturing semantic meaning
- Two common forms
  - token-level embeddings (used for cross-attention)
  - pooled sentence embedding (global meaning of the prompt)

IMAGE REPRESENTATION

Step 2: Image (or latent) representation

Instead of working directly with pixels, most systems operate in a compressed latent space

$$\text{image} \xrightarrow{\text{Encoder}} z$$

- Benefits of latent space
  - lower dimensionality
  - faster training and sampling
  - easier modelling of global structure
- The generative model learns to transform random noise into a latent image representation, which is later decoded into pixels

WHERE IS CONDITIONING APPLIED

Conditioning is injected into the denoising or generation network so that text influences every stage of image formation

- Conditioning is applied
  - at multiple network layers
  - at multiple spatial resolutions
- The most common mechanism is **cross-attention**, which links visual features with linguistic features

CROSS-ATTENTION

At each spatial location of the image feature map, the model asks

"Which words in the prompt are relevant for this part of the image?"

- Mathematically
  - Image features act as queries Q
  - Text embeddings acts as keys K and values V

$$\text{Attention}\,(Q, K, V) = \text{softmax}(\frac{\text{QK}^{\text{T}}}{\sqrt{\text{d}}})\text{V}$$

- This allows object tokens to guide object shapes, attribute tokens to guide colour and texture, and relational tokens to influence spatial layout
- Hence, **cross-attention makes image generation prompt-sensitive** by linking textual meaning to visual structure and appearance

CONDITIONING IN DIFFUSION MODELS

In diffusion models, images are generated by gradually removing noise, while conditioning steers the denoising direction

- **Unconditional diffusion**
  - The model starts from noise and iteratively denoise

$$z_T \rightarrow z_{T-1} \rightarrow \cdots \rightarrow z_0$$

  - At each step, the model predicts what part of the signal should be restored

- **Conditional diffusion with text**
  - Each denoising step is guided by text embeddings

$$p(z_{t-1} | z_t, \text{text})$$

  - At every step, noise is removed and structure consistent with the text is added
  - This ensures the final image aligns with the prompt, not just visually realistic

GUIDANCE STRENGTH

Even with conditioning, the model must balance prompt alignment and visual realism. This balance is controlled by **guidance strength**

Classifier-free guidance mixes unconditional prediction and conditional prediction

$$\hat{\epsilon} = (1 + s)\,\epsilon_{\text{cond}} - s\epsilon_{\text{uncond}}$$

where

- s is guidance scale
- Higher $s \rightarrow$ stronger adherence to text
- Lower $s \rightarrow$ more diversity, weaker prompt control

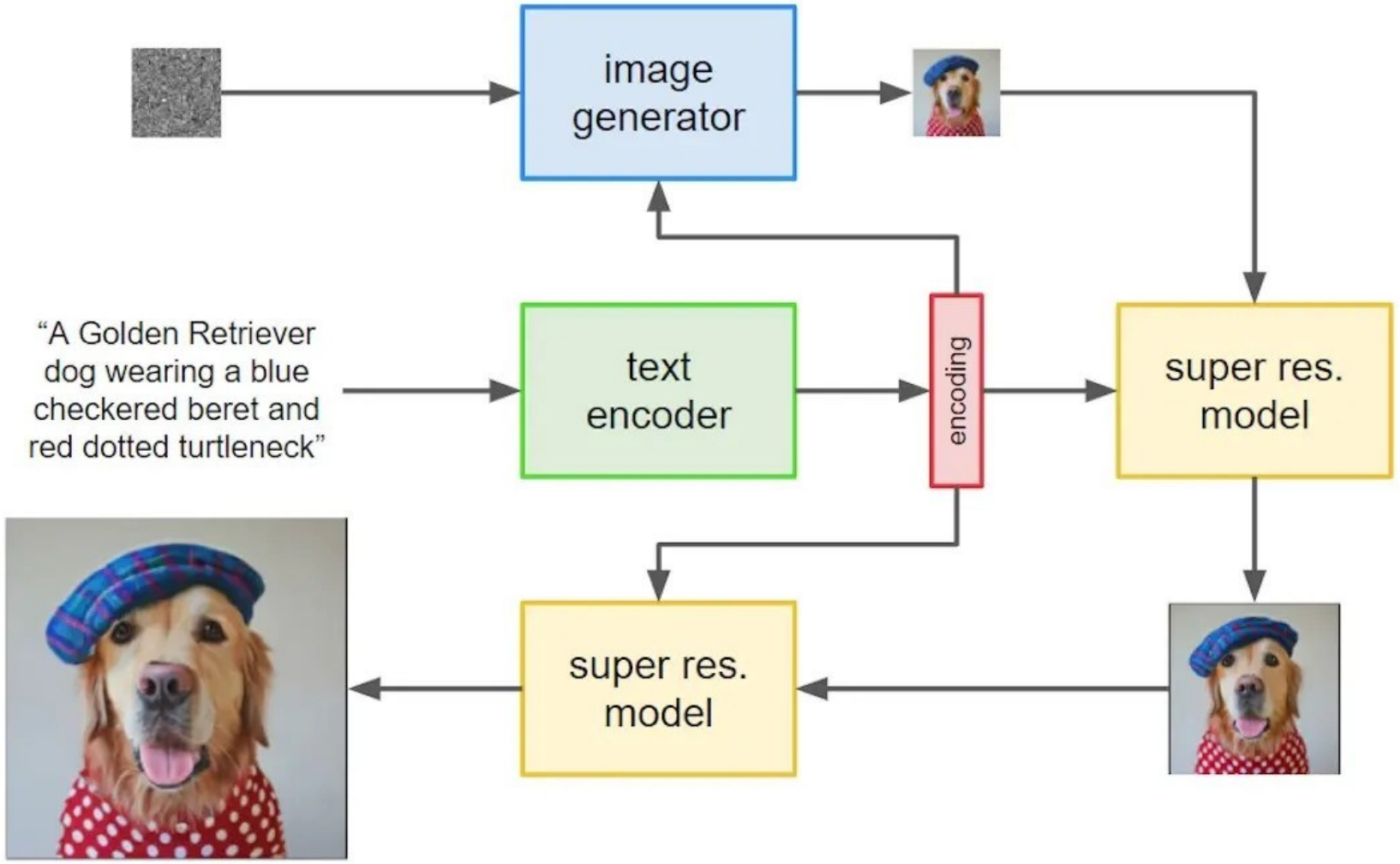
This is why prompts can be followed *loosely* or *strictly* depending on settings

WHAT TEXT CONDITIONING CONTROLS

Text conditioning can influence multiple semantic and visual factors simultaneously

Level	Controlled properties	Examples
Object presence	What entities appear in the image	“a dog”, “two cars”
Object attributes	Colour, size, material, shape	“red”, “wooden”, “shiny”
Style	Artistic medium, realism, illustration type	“watercolour”, “sketch”, “oil painting”
Scene context	Indoor/outdoor, time of day, environment	“on a beach”, “at night”
Relationships	Relative positions and interactions	“a cup on a table”, “person riding a bike”

TEXT-TO-IMAGE CONDITIONING ARCHITECTURE



OTHER CONDITIONING TYPES IN GENERATIVE MODELS

Text is only one possible source of conditioning

- In general, generative models learn

$$p(x \mid c)$$

where the condition c can come from different modalities or structural signals

- The core generative backbone (U-Net or transformer) usually remains the same
- What changes is how the conditioning signal is **encoded and injected** into the network

IMAGE-TO-IMAGE CONDITIONING

Here, the condition is another image

$$p(x_{\text{target}} \mid x_{\text{source}})$$

- Examples of image signals
  - super-resolution
  - image inpainting and outpainting
  - style transfer and domain translation
- The input image is encoded into latent feature maps and these features are provided to the denoising network as additional channels or feature-level conditioning signals
- This results in stronger spatial correspondence between input and output, where the structure is preserved and appearance is modified
- Compared to text conditioning, this provides **stronger geometric control but weaker semantic flexibility**

STRUCTURE OR LAYOUT-BASED CONDITIONING

Here, the condition specifies spatial constraints explicitly

$$p(x \mid \text{structure})$$

- Example of structural signals
  - Edge maps
  - Segmentation masks
  - Depth maps or bounding boxes
- Structure is encoded as spatial feature maps and injected into multiple U-Net layers, sometimes through dedicated control branches (not only attention)
- This enforces object placement and shape, and reduces ambiguity in composition
- This form of conditioning is used when **precise layout control is required**, which text alone cannot reliably provide

MULTI-MODAL CONDITIONING

Here, multiple conditions are provided together

$$p(x \mid c_1, c_2, \ldots, c_k)$$

- Example of multi-modal signals
  - Text + reference image
  - Text + pose map
  - Prompt + depth + edges
- Separate encoders are used for each modality and feature fusion is done using cross-attention and/or feature concatenation
- This combines semantic guidance from text with spatial or stylistic constraints from images or structures
- Modern controllable generation systems rely heavily on multi-modal conditioning

ARCHITECTURAL COMPARISON OF CONDITIONING MODELS

- **What usually remains the same**
  - backbone (U-Net or transformers)
  - Diffusion or decoding process
  - Iterative refinement over time steps

- **What usually changes**
  - Type of conditioning encoder
    - Text encoder
    - Image encoder
    - Structure encoder
  - How conditioning features are injected
    - Attention-based
    - Feature concatenation
    - Control branches

Thus, different conditioning tasks are variations of the same conditional generation framework, not entirely new model families

PROMPT SENSITIVITY

- Prompt sensitivity refers to the fact that small changes in the text prompt can lead to noticeable changes in the generated image, even when the overall meaning appears similar
- This occurs because the model conditions image generation on token-level embeddings, not only on overall sentence meaning
- Each token contributes separately to attention and feature modulation during image generation

WHY PROMPTS ARE SENSITIVE

- Cross-attention links image feature queries with text token keys and values at every denoising step
- Different words therefore compete for influence over visual regions, and emphasis shifts when token order or phrasing changes
- As a result, attributes may strengthen or weaken, and object focus may shift across the image
- Text-to-image models
  - do not understand the intent
  - learn statistical associations between text and images
  - optimise visual plausibility, not semantic correctness
- Due to these issues, prompts act more as soft constraints than strict instructions

PROMPT AMBIGUITY

- Prompt ambiguity occurs when a prompt can be interpreted in multiple valid ways. The model chooses the most statistically common interpretation, not the intended one

- Examples of ambiguous prompts

Prompt	Possible Interpretations
“I saw a man with a telescope”	Man holding telescope or being observed
“A painting of a crane”	Bird or construction crane
“A woman in a jacket”	Any style, age, or context

- Prompt ambiguity breaks generation as
  - Cross-attention distributes attention across multiple meanings
  - Competing interpretations weaken semantic alignment
  - The model collapses to a dominant visual pattern

PROMPT BIAS

- Prompt bias occurs when the generated image reflects stereotypes or skewed representations learned from data. Even neutral prompts can produce biased outputs
- Examples of prompt bias

Prompt	Common Bias
“A CEO”	Male, white, middle-aged
“A criminal”	Racial stereotypes
“A beautiful person”	Narrow beauty standards

- Prompt bias appears due to
  - Training data reflects real-world imbalances
  - Majority patterns dominate attention
  - Rare or counter-stereotypical examples are underrepresented
- Bias in generated images shapes perception, reinforces stereotypes and affects trust in AI systems

COMPOSITION LIMITS

- Compositionality means combining multiple concepts, attributes, and relationships correctly. Humans do this naturally, models often do not
- Examples of compositional failures

Prompt	Typical Failure
“A red cube on a blue sphere”	Colours swapped
“Three apples on a plate”	Wrong count
“A small elephant next to a big mouse”	Size incorrect

- Why composition is hard
  - Cross-attention competes for influence
  - Spatial reasoning is implicit, not explicit
  - Models lack symbolic reasoning
- The model knows what things look like, not how they relate